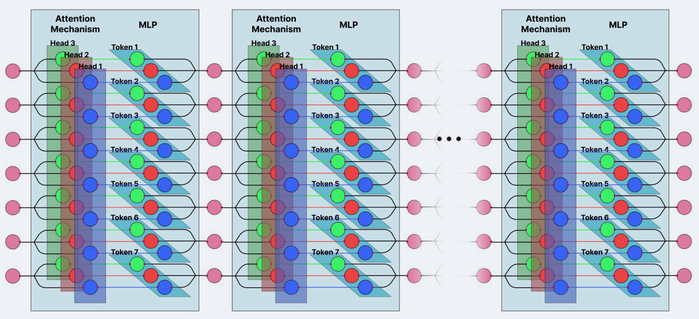
LIG: Layer-wise Integrated Gradients for Within-Layer Flow Analysis in Transformers

Eight Suzuki[1,3], Hideitsu Hino[1,2], Noboru Murata[1] [1]Waseda University [2]The Institute of Statistical Mathematics [3]Fujitsu Limited ✉ suzuki8@akane.waseda.jp

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Problem

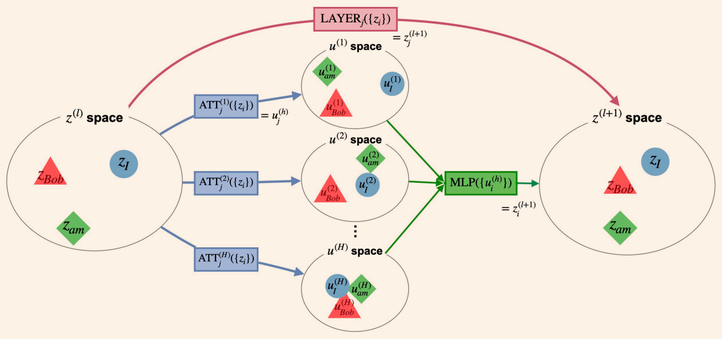
Explaining Transformers: what happens inside a layer?



- **Attention maps / rollout** [7, 5]: weights only
- **Norm-based** [6]: no ATT / MLP split
- **AttnLRP** [4]: a rule per operation
- **IG** [1]: scalar-output attribution

Idea OURS

LIG: compose attention and MLP scores.



$$\widetilde{\mathrm{IG}}^{\mathrm{LAYER}}_{i,j} \overset{?}{\approx} \sum_h \widetilde{\mathrm{IG}}^{\mathrm{ATT}}_{i,j,h}\, \widetilde{\mathrm{IG}}^{\mathrm{MLP}}_{h,j}$$

whole layer · composed

Set-to-set IG at the ATT and MLP boundaries, composed LRP-style, checked against whole-layer IG (≈?). One rule, no retraining.

ATT: tokens → heads · MLP: heads → output tokens.

Read on: §2 how · §4 numbers · §3 example

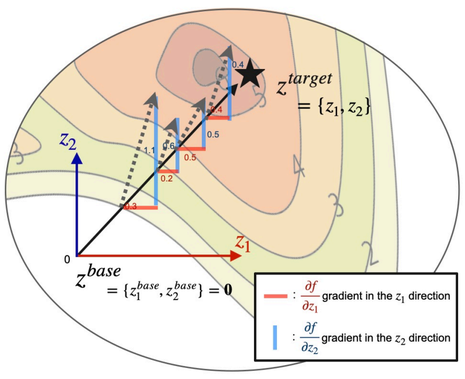
Result

Closest agreement: baselines matched to the whole layer.

ITB: ATT starts at the target token; MLP at the ATT output there. Zero: both at zero.

> Small margins; within each reference block, among tested configurations. BERT-base / PTB (§4); attribution agreement, not model behaviour; residual route omitted.

1 IG and its baseline



PRIOR IG [1] integrates gradients along the straight path $z(a) = z(0) + a\,(z - z(0))$ from a **baseline** $z(0)$ to the input z:

$$\mathrm{IG}_d = (z_d - z_d(0)) \int_0^1 \frac{\partial f(z(a))}{\partial z_d}\, da$$

The baseline sets what each contribution is measured against.

> **Contribution** = displacement × path-averaged gradient; $f(a) = \|y_j(a) - y_j(0)\|_2$, $\sum_r \mathrm{IG}_r = f(1)$.

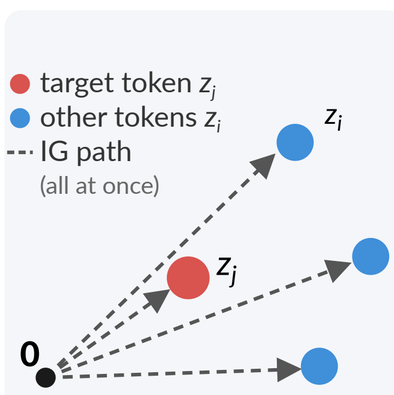
2 LIG: IG at the ATT and MLP boundaries OURS

Set-to-set IG (not standard in prior IG applications) attributes output token j's change (L_2, §5) to input tokens (ATT) or heads (MLP):

ATT	MLP (W_o, LayerNorm, FFN; residual z fixed)	Whole layer l
$z_i \to u_j^{(h)}$: $\mathrm{IG}^{\mathrm{ATT}}_{i,j,h}$	$u_j^{(h)} \to z_j^{(l+1)}$: $\mathrm{IG}^{\mathrm{MLP}}_{h,j}$	$z_i \to z_j^{(l+1)}$: $\mathrm{IG}^{\mathrm{LAYER}}_{i,j}$

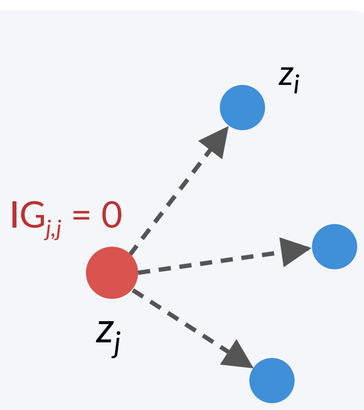
Baseline = start ($a = 0$); all inputs move together to their actual values ($a = 1$).

Zero



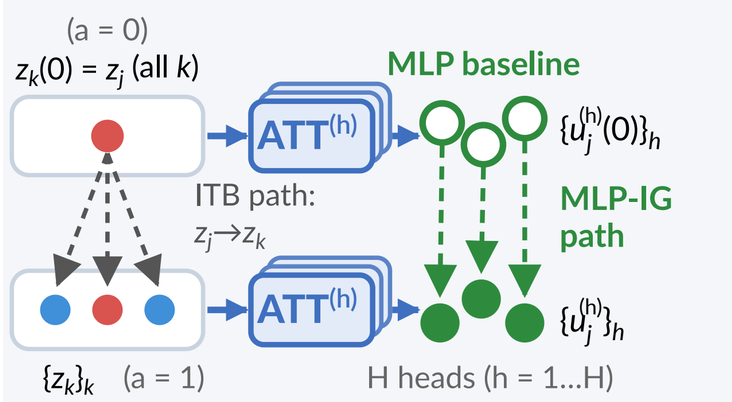
Zero: all tokens start at $\mathbf{0}$.

Input Token Base (ITB)



ITB: all tokens start at the target z_j, whose own contribution is zero by construction [8].

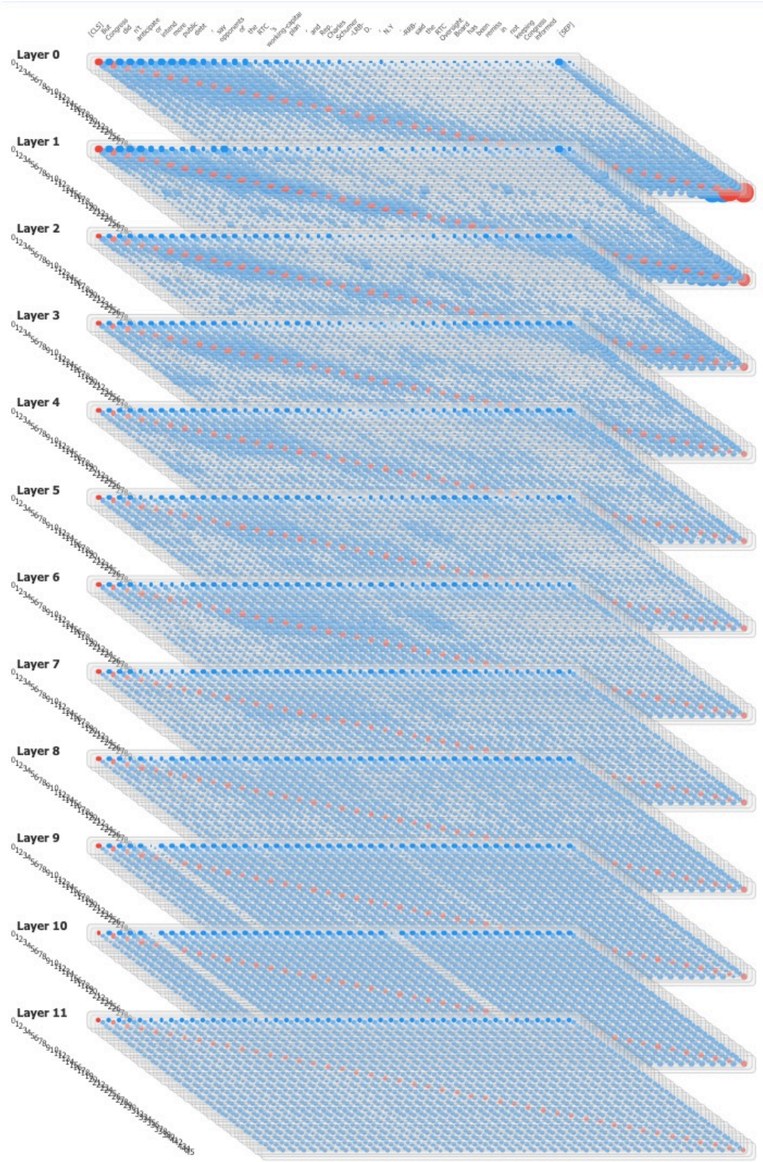
MLP: ATTITBa=0



ATTITBa=0: the MLP starts at each head's ATT output at the ITB start (all tokens $= z_j$).

3 Example: whole-layer IG maps

Whole-layer IG (no ATT / MLP split) · Zero baseline · one PTB sentence (qualitative). Columns: inputs; rows: outputs; size: |contribution|; red: self; blue: others. Here, layer 0 is near-diagonal; layers 9–11 span the sentence.



4 Result: composition vs. whole-layer IG

Compositional gap: L_2 distance between whole-layer and composed attribution vectors (over input tokens; per sentence, layer, output token). The LRP-style composition [2, 3] is **not an IG chain rule** (no residual route), so the two need not agree.

Setup. BERT-base-uncased · 1,700 PTB dev sentences [9] · $m = 32$ steps. Top 3 per reference; **bold**: best; **compare only within blocks.**

Whole-layer baseline (reference)	Composed: ATT	MLP baseline	$L_2 \downarrow$
ITB Input Token Base: all tokens start at z_j	**ITB**	**ATTITBa=0**	**0.305**
	ITB-zeroRatio	ATTITBa=0	0.312
	ITB-mapRatio	ATTITBa=0	0.318
Zero zero vector $\mathbf{0}$	**Zero**	**Zero**	**0.736**
	ITB-zeroRatio	Zero	0.744
	Zero	ATTITBa=0	0.749

ITB-zeroRatio / -mapRatio: ITB's zero self-score replaced by the Zero-baseline self-score / the attention weight $\alpha_{j,j}$; total preserved.

> **Best MLP baseline: ATTITBa=0 under ITB (for every ATT baseline tested); Zero under Zero.**

5 Score definition

Per input token or head r: sum IG over its dimensions, then normalize by the total change (signed):

$$\mathrm{IG}_{r,j} = \sum_{d \in r} \mathrm{IG}_d, \qquad \widetilde{\mathrm{IG}}_{r,j} = \frac{\mathrm{IG}_{r,j}}{\|y_j(1) - y_j(0)\|_2}$$

6 Limits & next steps

- **Limits:** BERT-base / PTB only; straight paths may leave the data distribution; the ITB completion may not be unique; L_2 scalarization discards direction.
- **Next:** decoder-only models; comparison with EAP-IG [10, 11]; intervention tests.

[1] Sundararajan+ ICML 2017 · [2] Bach+ PLOS ONE 2015 · [3] Montavon+ LRP overview 2019 · [4] Achtibat+ (AttnLRP) ICML 2024 · [5] Abnar & Zuidema ACL 2020 · [6] Kobayashi+ EMNLP 2020 · [7] Jain & Wallace NAACL 2019 · [8] Sturmfels+ Distill 2020 · [9] Marcus+ Treebank-3, LDC 1999 · [10] Syed+ BlackboxNLP 2024 · [11] Hanna+ COLM 2024.
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